

A BRIEFING FOR VINH LE · FROM STEPHEN SOOKRA

PIT WALL

*The post-race AI race engineer for the
drivers professional motorsport forgot —
and how the two of us are going to ship it
in twelve days.*

CHALLENGE

IBM SkillsBuild AI Builders
"AI Beyond the Finish Line"

STACK

IBM Granite · 8 models
Langflow · IBM Bob

SUBMISSION

May 31, 2026 · 11:59 PM ET

TARGET OUTCOME

Top 3 · Best Use of Tech
Most Innovative · Grand Prize

NAVIGATE THIS DOCUMENT

What's *inside*.

Read this front-to-back the first time. After that, treat it as a reference — anything you forget about the architecture, our lanes, or the Q&A defense is probably in here.

01	The Problem	03
	Why we're building this and who it's for	
02	What We're Building	04
	PIT WALL explained like you've never heard of it	
03	The Hidden Trap	06
	Kinetic Hallucination and the PhysicsTTM mitigation	
04	The Stack	08
	Every IBM Granite model and what it does	
05	Your Lane	10
	Exactly what Vinh owns on this build	
06	The 12-Day Plan	12
	Day-by-day, who does what	
07	Q&A Defense	14
	Five rehearsed answers for May 31	
08	After May 31	16
	The galaxy-tier ambition — NeurIPS, IBM, real product	

READ ME FIRST

Everything in this brief came from ~30 days of structured research — six AI models pressure-testing the idea, a NotebookLM verdict, and a Sookra Methodology v3.3 pass. You don't need to know any of that. Just read straight through.

CHAPTER ONE • THE GAP NOBODY IS FILLING

Every F1 driver has a *race engineer*. Most of motorsport doesn't.

A professional race engineer costs *£400–£500 per day*. That price tag is the reason ninety-five percent of motorsport runs without one — and the reason the drivers who need coaching the most can't get it.

£500/day

COST OF A PRO RACE COACH
(DRIVER61 PUBLISHED RATE)

2017

YEAR THE FIA FINALLY
LIFTED THE BAN
ON DISABLED SINGLE-SEATER
DRIVERS

400M

FANS IBM SHIPS POST-RACE
GRANITE
AI SUMMARIES TO • FOR
FERRARI

Who we're building for

Three groups, one problem:

- **Disabled racing drivers** running adapted controls — Team BRIT in the UK, Resilience Racing in the US, FFSA Handikart in France. They drive cars built around hand-control hardware and pneumatic-electronic shifters. Existing AI coaches assume able-bodied physics and systematically misread their telemetry.
- **Veteran-recovery teams** — Mission Motorsport in the UK, Operation Motorsport in the US. They use motorsport as a structured environment for medically-releasing service members. They have access to the cars; they don't have race engineers.
- **Grassroots racers** — anyone who's saved up for a Radical or a track-day BMW. They get the car, they get the seat time, they don't get the coaching layer that turns seat time into lap time.

Why IBM cares

IBM ships post-race AI race summaries to Scuderia Ferrari's ~400 million fans using their watsonx stack — Granite models, IBM Bob, watsonx.governance. **It's the exact same architecture** we're going to deploy. They built it for the Tifosi. We're pointing it at the disabled veteran in a hand-controlled BMW who has nobody on his pit wall.

« The same Granite stack that serves 400 million Ferrari fans — re-pointed at the drivers who need a race engineer most. »

Why nobody else is here

Track Titan raised \$5M in December 2025 to coach sim racers. **Trophai.ai** sells subscription coaching for iRacing. **Formula E + Google Cloud** built a "Driver Agent" in March 2025 — but it's sim-only and Gemini-based. None of them touch real-world paddock telemetry. **None of them ingest the FIA Certificate of Adaptations.** None of them know that a hand-control driver can press throttle and brake at the same time — and when they see that pattern, they flag it as a driver error.

That last sentence is the whole project in one line. Existing AI race coaches are *physically hostile* to adaptive drivers. We're building the only one that isn't.

THE DEEPEST INSIGHT

Track Titan, Trophi.ai, Formula E's Driver Agent — they all assume the driver's two feet move independently of each other. For a Team BRIT driver running a single pneumatic-electronic hand-control unit, that assumption breaks at the physics level. That's not a feature gap. That's an architectural mismatch.

CHAPTER TWO • THE PRODUCT IN PLAIN ENGLISH

PIT WALL, *explained simply.*

A driver finishes a session. They drop three files into our web app. Sixty seconds later they get back a corner-by-corner coaching report, a specific tuning recommendation, and a safety-verified stamp. That's the whole product.

The three inputs

INPUT	WHAT IT IS	WHERE IT COMES FROM
Telemetry CSV	The numerical log from the car — speed, throttle, brake, steering, lateral G, longitudinal G, RPM, gear, plus hand-control channels for adaptive drivers	MoTeC, AiM, RaceCapture — whatever data logger the team uses. We can also build from public FastF1 exports for testing.
FIA Certificate of Adaptations (PDF)	The official document that says exactly what hardware adaptations the driver runs — lever travel, paddle force, brake bias, simultaneity permissions	FIA Disability and Accessibility Commission. Every disabled FIA-licensed driver has one.
Written debrief (text)	The driver's own notes from the session: "Turn 4 felt loose," "couldn't get on power out of the chicane"	Driver types it. Free-form.

The four outputs

- A **corner-by-corner coaching report** in the voice of a race engineer ("Turn 4: your lever brake input averaged 70ms longer than your reference lap. Reduce lever travel by 4mm.")
- A **tuning recommendation** — a specific delta to apply for the next session
- A **next-session forecast chart** — what the lap will look like under the proposed change
- A **safety-verified stamp** — confirmation that the recommendation stays inside the FIA Certificate of Adaptations envelope and inside the laws of physics

The data flow, end to end

This is the simplest version of what the system does, in order. Memorize this. If anyone asks how PIT WALL works, you can walk them through these steps.

1	Granite-Docling 258M	Reads the FIA Certificate of Adaptations PDF. Outputs structured JSON with every adaptation parameter.
2	Granite Vision 4.1 4B	Reads the official timing sheet PDF. Outputs lap times and sector splits as CSV.
3	Granite TimeSeries TTM r2.1	Takes the telemetry CSV. Forecasts the next 1–2 laps under the proposed adaptation changes. This is the hero technical move.
4	PhysicsTTM projection	Clamps the TTM forecast onto a physically valid manifold. Prevents the model from forecasting impossible physics.
5	Granite 4.1 8B Instruct	Takes the corrected forecast + COA JSON + driver debrief. Writes the natural-language coaching report in race-engineer voice.
6	Granite Guardian 4.1 8B	Audits the physics-violation log via Bring-Your-Own-Criteria. Applies the safety-verified stamp.
7	Langflow + IBM Bob	Langflow is the visible orchestration the judge sees on screen. Bob is the agent that built the UI — same Bob IBM cites in the Ferrari case study.

What makes this technically novel

The IBM Granite TimeSeries TTM model was pretrained on *energy grids, weather, and retail sales data*. It has never been applied to motorsport telemetry in any published paper, IBM cookbook, Hugging Face Space, or hackathon project we can find. **To the best of our knowledge, PIT WALL is the first.**

That alone is a strong "Best Use of Technology" claim. But the deeper novelty is that we don't just feed the model telemetry — we wrap it with a physics-projection layer that prevents the kind of "kinetic hallucinations" foundation models are known for. That's the topic of Chapter 3.

CHAPTER THREE · THE OBJECTION THAT ALMOST KILLED US

Kinetic *Hallucination*.

When we ran the project through NotebookLM as a final validation pass, it surfaced a structural objection no other model had caught. Before we knew about it, PIT WALL had a fatal Q&A flaw. Now it's our deepest defense.

The problem in plain language

A race car obeys Newton. Speed is the integral of acceleration. Lateral G is a function of speed and steering angle. Throttle at zero means you can't accelerate. Tires have a finite grip ceiling. These are not negotiable laws.

Granite TimeSeries TTM, on the other hand, was trained on weather data. It has no idea what Newton is. When it forecasts eight telemetry channels at once, it forecasts each one *independently* — speed without knowing about throttle, lateral G without knowing about steering.

THE KILLER Q&A QUESTION

"Without a physics-informed layer, what stops your model from forecasting a lap where the car pulls 4G of lateral load with zero steering input, or where speed increases while the throttle is at zero?"

If we ship without a fix, a motorsport judge looking at our forecast on screen can spot impossible physics in seconds. Game over. **Without a mitigation, the whole project collapses on Q&A.**

The fix: PhysicsTTM, three layers

This is the architecture we landed on after running the problem through Claude's research tool. It's the single most defensible mitigation in 12 days.

A	Frozen TTM forecast	Granite TimeSeries TTM r2.1 produces the multivariate forecast as before. We don't retrain it. It stays frozen.
B	Physics projection	Every forecast step passes through a small optimization layer that clamps the output onto a physically feasible manifold — friction

The non-obvious part: brake-throttle simultaneity

This is the deepest piece of the whole project. Get this right and we have an answer that makes incumbents look broken.

Standard race-engineering physics says `throttle × brake = 0`. You can't press both pedals at the same time, because in an able-bodied car you physically can't reach both at once.

This rule is wrong for adaptive drivers. Team BRIT and MME Motorsport's electronic hand-control system explicitly supports holding full throttle while applying braking mid-corner — Professional Motorsport World has them on the record saying so. Adaptive drivers *need* this capability. It's not a bug, it's the adaptation.

So our physics-projection layer reads a flag from the FIA Certificate of Adaptations. When the COA says "simultaneity permitted," the constraint `throttle × brake = 0` is dropped. When it says "not permitted," the constraint stays. The same model serves both populations correctly.

« Track Titan, Trophi.ai, Formula E's Driver Agent — they all use able-bodied physics. PIT WALL is the only AI race engineer that reads the COA as a flag at the tensor level. »

What this opens up

This insight is the deepest "first" in the project. When a judge asks "why can't a Team BRIT driver just use Track Titan?", we have the knockout answer: *Because Track Titan will diagnose every adaptive driver as making a mistake every time they brake and throttle together.*

That's not a feature gap. That's a fundamental architectural mismatch. And we're the only project that knows it.

What's still imperfect

We're going to be honest with the judges about three things PhysicsTTM doesn't fully solve:

~~Constant μ friction circle. Real tire dynamics follow the Pacejka magic formula — a "heart-shaped" plot of~~
PIT WALL • A BRIEFING FOR VINH
~~Circle. It uses a constant 12 g peak acceleration, is flat, asym-~~

→ **1-Hz aggregation gap.** We downsample 50Hz raw telemetry to 1Hz to fit TTM's pretraining envelope.

CHAPTER FOUR • EVERYTHING WE'RE USING AND WHY

Eight IBM tools. *One pipeline.*

Every piece of the stack is an IBM Granite model used in its documented sweet spot. Nothing is bolted on. This is what wins "Best Use of Technology."

GRANITE-DOCLING 258M — DOCUMENT PARSER

Reads the FIA Certificate of Adaptations PDF and outputs structured JSON. Tiny (258M parameters), open-source, runs on CPU.

GRANITE VISION 4.1 4B — TABLE/CHART PARSER

Reads the official timing sheet PDF (sector times, lap charts) and outputs CSV. Used in its documented sweet spot — structured PDFs, not phone photos.

GRANITE TIMESERIES TTM R2.1 — FORECASTER

Sub-1M parameter time-series foundation model. Takes the telemetry CSV, produces the next-lap forecast tensor. NeurIPS 2024. Runs on CPU. **The novel application: no published prior art on TSFMs in motorsport telemetry.**

PHYSICSTTM PROJECTION LAYER — CUSTOM CODE (THIS IS MOSTLY VINH'S LANE)

Differentiable optimization layer that clamps TTM's forecast onto a physically valid manifold. Built with CvxpyLayers + NumPy. ~600 lines of original code. **This is the workshop-paper-worthy novel contribution.**

GRANITE 4.1 8B INSTRUCT — RACE-ENGINEER VOICE

Takes the corrected forecast + COA JSON + driver debrief. Writes the coaching report. Tone-engineered to read like a real F1 race engineer.

GRANITE GUARDIAN 4.1 8B — SAFETY AUDITOR

Reads a textual log of physics violations from the projection layer. Issues yes/no verdict via Bring-Your-Own-Criteria. Becomes the "Safety Verified" stamp.

LANGFLOW — VISIBLE ORCHESTRATION

The latency reality check

One of the six AI models we ran the project through called our original "60 second end-to-end" demo claim "fiction." They were right. On a laptop CPU, full document parsing alone can take 10–15 minutes.

The fix is the **PIT WALL Lite** demo architecture: we pre-cache document parsing at "driver onboarding" and only run the post-race coaching loop live. That coaching loop fits in 60 seconds on the RTX 4060 because every model after Docling/Vision is fast.

STEP	LIVE OR PRE-CACHED?	WHY
FIA COA PDF parse	Pre-cached	The COA doesn't change between sessions. A driver registers once.
Timing sheet parse	Pre-cached	Same reason. Onboarded once per event.
Telemetry ingest	Live	Fast. CSV parsing is sub-second.
TTM forecast	Live	Sub-1M parameters. Runs in < 1 second.
PhysicsTTM projection	Live	Small QP per step. Sub-millisecond.
Granite Instruct briefing	Live	~75 tok/s on RTX 4060. 300-word report = ~4 seconds.
Guardian audit	Live	Run in parallel with the briefing. ~1 second.

Total live demo: well under 60 seconds. When a judge asks "isn't 60 seconds impossible?" we'll honestly explain what's cached and why. That's the answer (see Q&A Card 5).

Hardware split

Stephen has the M2 MacBook. Vinh, you've got the heavier inference path on whatever your strongest machine is. The RTX 4060 laptop is where we run the actual demo recording. For development:

→ **M2 MacBook** — frontend (Streamlit / Bob-built UI), Langflow graph design, light testing

→ **Pi 4 / Pi 5** — stretch goal, post-hackathon edge demo

CHAPTER FIVE · VINH — THIS IS WHAT YOU OWN

You're *backend & physics*. I'm *frontend & pitch*.

Same split as Trace. Same split as every hackathon we've shipped together. The reason it works: you'd rather solve the math than rehearse the pitch, and I'd rather rehearse the pitch than solve the math. PIT WALL is no different.

COMPONENT	OWNER	WHAT IT MEANS IN PRACTICE
Granite TimeSeries TTM integration	VINH	Wire up the TTM inference path. Zero-shot first, then fine-tune on FastF1 telemetry with <code>decoder_mode="mix_channel"</code> . Output a clean forecast tensor.
PhysicsTTM projection layer	VINH (LEAD)	The friction-ellipse + bicycle-model + COA-flag projection. V1 in pure NumPy. V2 CvxpyLayer if time permits. ~600 LOC. This is the original technical contribution.
Physics-violation log serializer	VINH	The Python function that turns a numerical violation into a text log Guardian can read. Unit-tested for every violation type — this is safety-critical code.
Granite Guardian BYOC integration	VINH	Wire the text log into Guardian 4.1 8B with the custom criterion. Return the yes/no verdict.
Granite-Docling / Vision integration	VINH	Parse the FIA COA PDF and timing sheets. Output cached JSON. Backend-only; not in demo critical path.

Stephen's lane (for context)

COMPONENT	OWNER	WHAT IT MEANS IN PRACTICE
Granite Instruct briefing prompt	STEPHEN	Prompt-engineer the "race engineer voice." Iterate against real driver debrief examples.
Langflow orchestration graph	STEPHEN (WITH VINH ON BACKEND WIRING)	Visual graph the judge sees. Stephen handles layout; Vinh handles the data flow.
Frontend (React + shadcn or Streamlit)	STEPHEN	Drop-zone for telemetry CSV. "Run Race Debrief" button. Output panel: corner-by-corner report, TTM forecast chart, Guardian stamp, g-g diagram.
g-g diagram visualization	STEPHEN	Side-by-side raw TTM vs physics-corrected scatter. Friction-circle overlay. This is the demo's hero visual.
3-minute pitch video	STEPHEN	Script, screen-record, voiceover. Lock by Day 11.
Stakeholder outreach & submission	STEPHEN	Phase 1 emails sent. Phase 2 emails sent tonight. DoraHacks/Devpost submission Day 12.

THE LANE PRINCIPLE

The split is based on where each of us is fastest. You ship working backend in your sleep. I rehearse pitches in my sleep. Don't help me with the pitch unless I ask. I won't help you with the projection layer unless you ask. We sync at 9 PM and trust each other in between.

Your tomorrow morning, in three steps

I'm not asking you to read the whole report and figure out where to start. Here's exactly what to do on Day 1:

STEP

1

Stand up the GitHub repo

Create `github.com/StephenSook/pit-wall` or `/physics-tsfm`. Apache 2.0 license. Empty README. Add me as a collaborator.

VINH – 15 MINUTES

STEP

2

Get TTM running zero-shot on a FastF1 slice

`pip install granite-tsfm`. Pull `ibm-granite/granite-timeseries-ttm-r2`. Run zero-shot inference on any 5-lap FastF1 export. Just confirm the inference path produces a forecast tensor without errors.

VINH – HALF A DAY

STEP

3

Sketch the physics-validator function signature

No implementation yet. Just write the Python function signatures: `friction_ellipse_check()`, `forward_euler_consistency()`, `bicycle_kinematic_check()`, `coa_simultaneity_rule()`. We'll discuss together before you implement.

VINH – 30 MINUTES

How we work together

- **Daily 15-minute sync.** 9 PM ET. Discord. Show your screen, show what's working, show what's broken. No meeting agenda needed.
- **Commit early, commit often.** Even broken code goes in a branch. I want to see progress in the repo.
- **Don't optimize for cleverness, optimize for shipping.** V1 NumPy beats V2 CvxpyLayer if it ships by Day 6. We can always upgrade.
- **If you're blocked, message immediately.** Don't waste an evening on a problem that's a five-minute conversation.
- **If something I asked for is dumb, push back.** You've shipped backend on Trace, on Saber, on PyroLens. Your instinct is good. Use it.

CHAPTER SIX • FROM TODAY TO SUBMISSION

Twelve days. *Every one of them counted.*

Today is May 19. Submission is May 31, 11:59 PM ET. We're shipping on May 30 to leave a 24-hour buffer.

MAY 20

1

Repo + TTM zero-shot path

GitHub stood up. TTM zero-shot inference verified on a FastF1 telemetry slice. Stephen confirms Langflow + Granite 4.1 8B Instruct load on local hardware.

BOTH

MAY 21-22

2

Physics validator (V1, NumPy)

Vinh implements `friction_ellipse_check`, `forward_euler_consistency`, `bicycle_kinematic_check`, `coa_simultaneity_rule`. Wraps as `PhysicsViolationLog` dataclass with text serialization. Unit tests for every violation type.

VINH

MAY 23-24

3

Projection layer + Guardian wiring

Day 5: V1 NumPy projection of (a_long, a_lat) onto the ellipse boundary. Day 6: V2 CvxpyLayer QP if Day 5 is clean. Day 7 in parallel: Vinh wires the violation log into Granite Guardian 4.1 8B BYOC.

VINH

MAY 25

4

UI + g-g diagram

Stephen builds the Streamlit (or Bob-built React) UI. Drop zones, "Run Race Debrief" button, side-by-side g-g diagram with friction-circle overlay.

STEPHEN

MAY 26

5

TTM fine-tune + ablation

5% data fine-tune of TTM r2 with `decoder_mode="mix_channel"`. Backbone frozen. Three-way ablation: zero-shot / fine-tuned / fine-tuned + projection. Numbers go on the deck.

VINH

Days 6 through 9 — integration and ship

MAY 27

6

Docling + Vision (out of critical path)

Granite-Docling on FIA COA → JSON. Granite Vision on timing sheet → CSV. Pre-cached. Not in the live demo loop, but built for the production story.

VINH

MAY 28

7

Langflow + Bob orchestration

Full pipeline wired in Langflow: Docling → COA JSON → TTM → projector → Guardian → Instruct briefing → Vision g-g chart. Bob agent on top.

BOTH

MAY 29

8

Dress rehearsal + Q&A drills

End-to-end demo on RTX 4060. Time every step. Stephen runs Vinh through all five Q&A flashcards. Hostile pass with follow-ups.

BOTH

MAY 30

9

Submission — 24 hours early

3-minute demo video recorded. README polished. DoraHacks/Devpost submission complete. We do not push to the last 24 hours.

BOTH

Decision gates — when we cut features

Hackathons fail when teams refuse to cut. We're going to cut early and often if anything slips. Here are the explicit decision gates:

BY THIS DAY	IF THIS ISN'T WORKING	WE CUT
Day 4 (May 23)	Zero-shot TTM doesn't beat a seasonal-naive baseline on real telemetry	Pivot to fine-tune-first. Skip zero-shot story in the pitch.
Day 6 (May 25)	V2 CvxpyLayer projection has convergence issues	Ship V1 pure NumPy. Move CvxpyLayer to V2 / workshop paper.
Day 8 (May 27)	Docling or Vision parsing is unreliable on real PDFs	Pre-process to static JSON/CSV. Frame as "production onboarding flow." See PIT WALL Lite.
Day 10 (May 29)	Live demo doesn't fit in 60 seconds on the RTX 4060	Pre-record the demo. Use the live UI for Q&A only.
Day 11 (May 30)	Three or more components are unstable	Ship PIT WALL Lite end-to-end. We have a working product; we don't need a perfect one.

What we are explicitly not doing in 12 days

- **No real-time pit strategy.** Post-race only. "Beyond the finish line."
- **No 3D track modeling.** Constant- μ 2D friction circle. V2 work.
- **No full Pacejka tire model.** Heart-shape friction is V2 / paper.
- **No real-world telemetry partnership.** Public FastF1 + synthetic hand-control channels for now. Real data is post-hackathon.
- **No mobile app.** Web only.
- **No multi-driver comparison.** Single driver, single session.

Every one of these is a real product feature for the post-hackathon arc. They're not in the demo because the demo doesn't need them — and shipping a focused demo beats shipping an overstuffed one.

CHAPTER SEVEN • THE FIVE REHEARSED ANSWERS

Memorize *these five*.

Q&A is won by preparation, not improvisation. Five cards. Twenty to thirty seconds each. Both of us memorize all five. Whoever feels the question better takes the answer; the other nods.

CARD 01 – DEEP DYNAMICS**"Isn't this just Deep Dynamics with an IBM logo?"**

THE ATTACK

They already built a Physics Guard layer for autonomous race cars. What's actually new here?

THE ANSWER (30 SECONDS)

Deep Dynamics is a brilliant precedent, but it solves a different problem. They trained a bespoke neural network from scratch on race-car data. PIT WALL proves a different pattern: we take a frozen, general-purpose foundation model — Granite TTM, pretrained on weather and retail — and wrap it with a differentiable physics-projection layer at inference time. We aren't building a custom PINN; we're constraining an off-the-shelf TSFM. To our knowledge no prior published work does that for vehicle dynamics without retraining the model from the ground up.

CARD 02 – KINETIC HALLUCINATION**"How do you stop TTM forecasting impossible physics?"**

THE ATTACK

TTM was trained on energy and weather. How do you stop it forecasting 4G lateral with zero steering, or speed increasing with throttle at zero?

THE ANSWER (30 SECONDS)

That's exactly the problem we built PhysicsTTM to solve. After TTM produces the forecast tensor, every step passes through a differentiable QP layer that enforces three hard constraints: the friction ellipse — total acceleration bounded by $\mu \cdot g$ — a forward-Euler kinematic check tying speed to longitudinal acceleration, and the bicycle model tying lateral G to steering angle and speed. Lateral G with zero steering becomes infeasible by construction. We're honest about residual gaps: V1 uses constant- μ , not full Pacejka, and we acknowledge sub-second hallucinations inside 1-Hz aggregates. Those are V2 work. But the worst zero-order impossibilities are prevented before they reach the driver.

CARD 03 — SERIALIZATION GAP

"If Guardian audits text not tensors, how is that actually safe?"

THE ATTACK

Granite Guardian judges text, not numbers. If your Python serializer has a bug, your whole safety story collapses. How do you protect against that?

THE ANSWER (25 SECONDS)

You're right that Guardian audits text, not tensors — we're explicit about that in the architecture. The defense is a unit-test suite covering every violation type: each kinematic failure mode has a fixture that produces a known text log and a verified Guardian verdict. The serializer is treated as safety-critical code, not glue code. We also surface Guardian's reasoning trace in think-mode in the UI, so the audit isn't a black box. The textual layer is a deliberate design choice with deliberate test coverage.

CARD 04 — COA SIMULTANEITY (KNOCKOUT)

"Why can't a Team BRIT driver just use Track Titan?"

THE ATTACK

Track Titan exists, it's funded, it's proven, it has 200,000 users. Why isn't this a solved problem?

THE ANSWER (30 SECONDS) — THIS IS YOUR OFFENSIVE CARD

Because they'll misdiagnose every adaptive driver they touch. Track Titan and Trophi.ai use able-bodied physics — their models assume throttle times brake equals zero. Team BRIT and MME Motorsport's electronic hand-control system explicitly supports simultaneous brake and throttle mid-corner. That's not a bug, it's the adaptation. So when those tools see simultaneous inputs, they flag it as a driver error and penalize technique that's actually required. PIT WALL is the only AI race engineer that reads the FIA Certificate of Adaptations as a flag at the tensor level — when the COA says simultaneity is permitted, our physics model permits it. We're not coaching disabled drivers with the same model as everyone else. We're coaching them with the model that matches their car.

The final card

CARD 05 — LATENCY

"Sixty seconds end-to-end on a laptop CPU? That's fiction."

THE ATTACK

Granite-Docling alone takes 10–15 minutes on a CPU. Your 60-second claim is engineering theatre.

THE ANSWER (25 SECONDS)

You're right that a cold-start, end-to-end pipeline can't fit in 60 seconds — that would be dishonest to claim. What runs live in the 60-second demo is the core race-engineer loop: TTM forecast, physics projection, Guardian audit, Granite Instruct briefing. Document parsing — the FIA COA and timing sheet — runs once at driver onboarding and is cached. That's how production AI race engineering actually works: a driver's adaptation profile doesn't change between sessions. The 60 seconds is the post-race coaching loop, not the registration flow.

How to rehearse

Three passes before May 31:

- **Solo pass.** Read each card aloud once. Time yourself. Adjust until each is under 30 seconds.
- **Partner pass.** Ask each other the questions in random order. Answer cold. If you stumble, mark that card for more reps.
- **Hostile pass.** Ask the question, then push back on the answer with one follow-up. ("But what if the QP layer doesn't converge?") This trains you for the second-order question after the first.

Three hostile passes the night before the submission video. The cards are a starting position — the actual Q&A may move sideways. Rehearsed core answers are how you stay calm when it does.

THE Q&A PRINCIPLE

Whoever feels the question better takes the answer; the other nods. If a question doesn't match a card exactly, redirect to the closest card. Don't ad-lib novel technical claims under pressure — that's how good projects die in Q&A.

CHAPTER EIGHT • THE ARC BEYOND THE HACKATHON

Galaxy-tier *ambition*.

The 12-day build is the entry ticket. The real prize is what comes after — a NeurIPS 2026 Workshop paper, an IBM Consulting partnership, and the first production-grade AI race engineer for adaptive drivers.

The workshop paper

To our knowledge, no published work wraps a frozen pretrained time-series foundation model with a differentiable physics-projection layer for vehicle dynamics. We can submit to a NeurIPS 2026 Workshop in Sydney.

- **June 6, 2026** — NeurIPS 2026 workshop proposals deadline
- **July 11, 2026** — Workshop acceptance notifications
- **August 29, 2026** — Suggested paper submission deadline
- **December 11–12, 2026** — Sydney workshop presentation

Working title: "PhysicsTTM: Physics-Projection Layers for Frozen Time-Series Foundation Models, with an Application to Adaptive Motorsport." Both names on it.

The IBM Consulting trajectory

If we win the SkillsBuild challenge, the canonical entry path is hackathon win + a public open-source library (we'll release `physics-tsfm` as Apache 2.0). Grok's analysis identifies this as the highest-leverage post-hackathon move.

The real product

If a data partnership lands with Team BRIT or Mission Motorsport — Phase 1 emails are already in flight — we have something genuinely new: the first AI race engineer tuned for the population existing incumbents can't serve. Spinout conversation, not a hackathon project. But it doesn't exist without the next 12 days.

The June extension. The IBM SkillsBuild June challenge is FIFA World Cup themed. PhysicsTTM

FINAL WORD

Let's *ship*.

The drivers who need a race engineer most don't have one. We're going to fix that — and we're going to do it with the same Granite stack IBM ships to four hundred million Ferrari fans.

Day 1 starts tomorrow morning. Read this brief end-to-end first. When you're ready, ping me and we'll kick off the repo together.

— STEPHEN

ATLANTA · MAY 19, 2026 · 11:47 PM ET